

AI facial expression valence: reproducibility in healthy Japanese adults for digital environments

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Abstract

Introduction: Artificial intelligence (AI)-based facial expression analysis can estimate emotional valence rapidly and noninvasively without the need for human training. However, reproducibility data remain limited, particularly in East Asian populations, where facial expressions may be more subtle. Establishing reproducibility in this context is essential for evaluating whether AI-derived valence measures may be useful in digital environments and mental health-related monitoring. Therefore, this study had two aims: (1) to evaluate state-specific reproducibility for positive, negative, and neutral facial expressions, and (2) to examine short-term trait-like stability by integrating valence estimates across all emotional conditions.

Materials and methods: Healthy young adults in Japan ($N = 40$) completed two facial expression recording sessions separated by 30 min. Positive, neutral, and negative facial expression conditions were recorded using an AI-based system. Test–retest reproducibility was assessed using intraclass correlation coefficients (ICCs) and Bland–Altman analyses. To examine short-term trait-like stability across conditions, all condition-specific valence values were additionally integrated into a single dataset for each participant.

Results: The ICC values were 0.82, 0.61, and 0.05 for the positive, negative, and neutral conditions, respectively. Bland–Altman analyses indicated no notable systematic bias across sessions. When data from all conditions were combined, the integrated dataset yielded a high ICC of 0.94, indicating short-term trait-like stability.

Conclusions: AI-based emotional valence values showed good to moderate reproducibility for positive and negative expressions, whereas the neutral condition showed low ICC values, likely because of restricted score variability. Integrating valence data across conditions yielded a high ICC, indicating short-term trait-like stability at the individual level. These findings support the potential of AI-based valence as a low-burden measure for future research on emotional assessment in digital environments. However, further validation under ecologically valid real-world conditions is needed.

Keywords: *artificial intelligence, facial expression analysis, emotional valence, test–retest reliability, intraclass correlation coefficient, digital mental health*

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1. Introduction

Artificial intelligence (AI)-based facial expression analysis has become increasingly widespread in psychology, education, and well-being research [1]. Recent research has increasingly adopted dimensional approaches to emotion, particularly the valence–arousal model, which conceptualize emotions as varying along continuous dimensions [2–4]. In this framework, valence, ranging from pleasant to unpleasant, is a core dimension of emotional experience and widely used in psychological research [5]. Advances in AI technology have further enabled the estimation of dimensional indices such as valence from facial expressions, offering practical advantages as a rapid, contact-free assessment method that does not require extensive human training [4]. In affective research, valence-related constructs, such as positive and negative affect, can be assessed both as a momentary mood state (“state”) and as a more general characteristic (“trait”) [6]. Trait-like tendencies are generally considered more stable and may help explain individual differences in emotional style and reactivity [7, 8]. Therefore,

it is important to examine whether AI-derived valence can capture state-specific responses as well as short-term trait-like stability across conditions, particularly if such measures are to be used in practical settings such as education and mental health.

Cultural factors are also known to affect the accuracy of AI-based facial expression analysis, including culture-specific display rules and differences in emotion judgments [9]. For example, individuals in East Asian cultures have been reported to display more subtle emotional expressions and suppress clear negative facial expressions more often than those in Western cultures [10]. This tendency may reduce the detection of negative facial cues, which in turn, can affect the cross-population validity of facial expression recognition algorithms. Furthermore, algorithm performance may be affected by dataset and demographic biases, potentially reducing cross-population generalizability [11, 12]. These issues are important because they may lead to systematic misestimation

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of AI-derived valence when such systems are applied to populations whose display rules differ from those represented in the original training data. Accordingly, reproducibility in East Asian populations should be established before broader application in research and practice. Furthermore, the intraclass correlation coefficient (ICC), which is a commonly used index of reproducibility for continuous outcomes in clinical research, is known to depend strongly on score distributions and between-subject variability [13, 14]. Because AI-derived valence may ultimately be used to assess individuals, reproducibility studies in clinical practice contexts using the ICC are important. At the same time, such evaluations require careful attention to score dispersion and between-subject variability [14]. Based on these considerations, the present study aimed to examine the reproducibility of AI-derived valence values in healthy young adults in Japan. A previous study using the same experimental framework examined concurrent validity and κ -based agreement of Kokoro Sensor valence classification in healthy individuals [15], whereas the present study focuses on ICC-based reproducibility, absolute reliability, and short-term trait-like stability of continuous valence values. This study had two main aims: (1) to evaluate state-specific reproducibility for positive, negative, and neutral facial expressions, and (2) to examine short-term trait-like stability by integrating valence estimates across all emotional conditions. This integrated analysis was intended to assess whether relatively stable individual differences in overall valence patterns could be observed across repeated measurements over a short retest interval, rather than to infer enduring personality-level affect. Clarifying these measurement properties is considered important for evaluating whether AI-derived valence may serve as a low-burden indicator for emotional assessment in digital environments and mental health-related monitoring.

2. Materials and methods

2.1. Study design and ethical approval

The present study was conducted in a controlled laboratory environment between March and June 2024 using an instructional facial expression task within a test–retest reproducibility design. This study aimed to investigate the reproducibility of valence scores derived from AI-based facial expression analysis under three conditions: positive, neutral, and negative. Standardized devices, including a high-resolution webcam and a personal computer, were used to collect data from all participants. All procedures adhered to the ethical standards established by the Helsinki Declaration and approved by the ethics committees at the School of Allied Health Sciences, Kitasato University, Japan (approval No. 2023-032). Written informed consent was obtained from all participants before data collection.

2.2. Participants

The study participants were healthy young adults recruited from a university in Japan. The eligibility criteria were as follows: (1) age between 18 and 30 years, (2) no history of psychiatric or neurological disorders, (3) no facial muscular impairments, and (4) no current treatment for insomnia, stress, or fatigue.

Japan was selected as the study setting to represent East Asian contexts, which have been shown to differ in display rules from

American populations [9]. In addition, previous research has suggested differences between Western–Caucasian and East Asian basic facial expressions of emotion [16]. These characteristics were considered important for investigating the reproducibility and robustness of AI-derived valence values across diverse cultural contexts. The Japanese version of the Profile of Mood States Second Edition—Adult Short Form (POMS 2-SF) was administered to confirm that the participants' mood states were within the normal range before data collection. The POMS 2-SF provides age-adjusted standardized T-scores for mood-related subscales and the Total Mood Disturbance (TMD) index, with a normative mean of 50 and a standard deviation (SD) of 10; higher TMD scores indicate greater overall mood disturbance. The POMS 2-SF has been used in recent studies to assess mood states in Japanese samples [17]. Participants with T-scores greater than 2 SDs above the normative mean were excluded from the analysis.

2.3. Emotion detection and valence estimation

The Kokoro Sensor (version 1.2.2.0, CAC Inc., Tokyo, Japan) was used for AI-based facial expression analysis. This AI-based emotion estimation system analyzes facial expression data based on the Facial Action Coding System and Ekman's theory of basic emotions. The Kokoro Sensor has been used in studies involving clinical populations [18, 19]. The software system provides outputs for seven core emotion categories—anger, contempt, disgust, fear, joy, sadness, and surprise—as well as three additional emotion-related output categories—sentimentality, confusion, and neutral—and calculates a valence index, positive–neutral–negative, from the weighted probabilities of these outputs. In the present study, the valence score, ranging from –100 to 100, was used as a continuous outcome, with higher values indicating more positive affect. The detailed weighting algorithm used to compute valence is proprietary and therefore not publicly disclosed.

2.4. Experimental procedure

Each participant completed two facial expression recording sessions separated by approximately 30 min to evaluate test–retest reproducibility. During each session, participants were seated approximately 60 cm from the camera and instructed to maintain a natural posture and minimize head and body movement so that the recording conditions remained as constant as possible. Prior to the first recording session, participants completed a short familiarization procedure, during which they practiced producing the three target facial expressions—positive, neutral, and negative—for approximately 10 s each under the guidance of the researcher. The researcher then verified the camera framing and instructed the participants to modify their posture and face position as needed to ensure consistent alignment throughout the video recording. For data collection, participants were asked to mimic three types of facial expressions corresponding to positive, neutral, and negative valence conditions voluntarily for approximately 30 s each while facing the PC camera. The neutral condition was described as a neutral facial expression, the positive condition as a happy facial expression, and the negative condition using examples such as sad, frustrated, or disgusted facial expressions. No specific instruction regarding expression intensity was provided; participants were simply asked to produce and maintain each target facial expression for approximately 30

s. The expressions were posed rather than induced by external emotional stimuli. No emotional induction procedures, such as images, videos, or autobiographical recall, were used. The same three expression blocks were repeated in the same order after a 30-min interval in the second session.

2.5. Statistical analysis

2.5.1. State-specific reproducibility

For each 30-second recording, the first and last 5 s were excluded, and the mean over the remaining 20 s was used as a representative valence value for each condition. The initial 5 s were excluded to minimize the influence of brief delays before stable facial detection and output by the software after recording onset. The final 5 s were excluded because participants often anticipated the end of the trial as the time limit approached, which could reduce maintenance of the target facial expression. This procedure was intended to obtain a more stable segment that better reflected sustained expression during each condition. For each condition (positive, neutral, and negative), test–retest reproducibility was assessed using the ICC (3,1) based on a two-way mixed-effects model with a consistency definition, assuming a fixed measurement system (Kokoro Sensor, AI-based analysis) and random participants. Because the Kokoro Sensor is a standardized measurement system based on a fixed algorithm, the measurement system was considered fixed rather than random. Therefore, test–retest reproducibility was evaluated using the ICC (3,1), which estimates consistency across repeated measurements. The ICC values were interpreted according to the following classification: <0.50 = poor, 0.50 – 0.75 = moderate, 0.75 – 0.90 = good, and >0.90 = excellent reproducibility. Bland–Altman plots were used for each condition to examine absolute agreement and distribution patterns visually. In addition to ICCs and Bland–Altman plots, absolute reliability indices were calculated for each condition, including the within-subject SD, mean difference between sessions, and 95% limit of agreement (LOA). In the present two-session repeated-measures design, the standard error of measurement was numerically identical to the within-subject SD. These indices were used to complement ICC-based interpretation, particularly in the neutral condition, where between-subject variability was expected to be small, which may be attributed to the restricted range of valence values in that condition.

2.5.2. Short-term trait-like stability (combined dataset)

All three condition-specific datasets were combined into a single integrated dataset for each participant to examine short-term trait-like stability across conditions. This approach was based on psychometric perspectives that distinguish state-specific variation from the relatively more stable trait-like components [8, 20]. The ICC (3,1) for the integrated dataset was interpreted as an index of short-term trait-like stability across integrated emotional conditions.

2.5.3. Supplementary descriptive and condition-congruence analyses

To support interpretation of the instructed valence conditions, supplementary descriptive and condition-congruence analyses were conducted. Condition-specific descriptive statistics were

calculated for the positive, neutral, and negative conditions in each session. In addition, the expected ordering of valence across conditions, positive $>$ neutral $>$ negative, was examined descriptively for each participant within each session. As supplementary analyses, overall within-session differences across the three conditions were evaluated using Friedman tests, and pairwise comparisons were examined using two-sided Wilcoxon signed-rank tests. These analyses were not designed to evaluate emotion estimation accuracy, but rather to provide descriptive support for the condition congruence of the valence value.

2.5.4. Sex-stratified sensitivity analysis

As a supplementary sensitivity analysis, reproducibility was also examined separately for male and female participants. ICC (3,1) values and their 95% confidence intervals (CIs) were calculated within each sex subgroup for the positive, neutral, and negative conditions, as well as for the combined dataset used to examine short-term trait-like stability. These analyses were intended to explore whether the overall reproducibility pattern was broadly consistent across sexes. Given the limited subgroup sample sizes, the results were interpreted descriptively as sensitivity analyses rather than as formal tests of sex differences.

2.5.5. Software and statistical significance

All analyses were conducted using R version 4.3.1 (R Foundation for Statistical Computing, Vienna, Austria). ICCs were calculated using the psych package (version 2.4.3; Revelle, 2024) [21]. For each condition, datasets were imported from CSV files using read.csv, and all columns were converted to numeric values using lapply (dataset, as.numeric) to avoid unintended type handling. Test–retest reproducibility was then evaluated by applying the ICC function from the psych package to each dataset, and the ICC (3,1) estimates and their 95% CIs were extracted from the output to represent consistency under a two-way mixed-effects model. For each condition, ICC analyses used a wide-format dataset with two columns representing session 1 and session 2 mean valence values per participant ($k = 2$). Bland–Altman plots were produced using ggplot2 (version 3.5.1; Wickham, 2016) [22], with limits of agreement defined as the mean difference $\pm 1.96 \times \text{SD}$. Bland–Altman differences were computed as session 2 minus session 1. For the supplementary condition-congruence analyses, Friedman tests were conducted using the friedman.test function from the stats package, and pairwise comparisons were conducted using two-sided Wilcoxon signed-rank tests with the wilcox.test function from the stats package. Holm adjustment for multiple comparisons was applied using the p.adjust function. Unless otherwise specified, statistical significance was set at $p < 0.05$.

2.6. Sample size estimation

The ICC analyses were conducted on a fixed sample ($N = 40$) determined by the parent validity study; thus, no separate a priori power calculation was performed for the ICCs. Nonetheless, the precision of estimates was documented via 95% CIs. Given two repeated measurements per participant ($k = 2$), a sample of $N = 40$ typically yields acceptable precision for ICC estimates (i.e., 95% CI widths on the order of ≤ 0.20 for ICCs ≥ 0.75), which is consistent with common recommendations for reliability studies [23, 24].

3. Results

3.1. Participant characteristics

Forty healthy young adults (23 men and 17 women; mean age, 22.1 ± 1.5 years; age range, 21–27 years) participated in the present study. All participants scored within the normal range on the POMS 2-SF, indicating no marked mood disturbance prior to testing, with a mean TMD T-score of 46.2 ± 9.1 (range, 31–66). Therefore, all 40 participants were included in the analyses.

3.2. State-specific reproducibility

The ICCs (3,1) for each emotional condition are presented in **Table 1**. The ICCs were 0.82 (95% CI: 0.68–0.90) for the positive condition, 0.61 (95% CI: 0.38–0.77) for the negative condition, and 0.05 (95% CI: –0.26 to 0.35) for the neutral condition,

indicating good, moderate, and poor reproducibility, respectively. To characterize agreement further, Bland–Altman plots were constructed for each state-specific condition to illustrate absolute agreement between sessions and distribution patterns (**Figure 1A–C**). The mean differences between sessions were small in all conditions, suggesting no clear systematic bias. Absolute reliability indices are presented in **Table 2**. The within-subject SDs were 11.81 for the positive condition, 13.69 for the negative condition, and 8.22 for the neutral condition. The corresponding 95% LOAs were –37.03 to 28.47, –42.48 to 33.40, and –24.43 to 21.15, respectively. Notably, despite its very low ICC, the neutral condition showed the smallest within-subject SD and the narrowest LOA range among the three state-specific conditions. This pattern supports the interpretation that the low ICC in the neutral condition was mainly driven by restricted between-subject variability rather than by poor absolute agreement.

Table 1 • Test–retest reproducibility of AI-derived valence values across emotional conditions.

Condition	ICC (3,1)	95% CI	Interpretation
Positive	0.82	0.68 to 0.90	Good
Negative	0.61	0.38 to 0.77	Moderate
Neutral	0.05	–0.26 to 0.35	Poor
Combined	0.94	0.92 to 0.96	Excellent

Values represent intraclass correlation coefficients (ICCs (3,1)) and their 95% confidence intervals (CIs). Interpretation follows conventional thresholds: <0.50 = poor, 0.50–0.75 = moderate, 0.75–0.90 = good, and >0.90 = excellent reproducibility. ICC (3,1): Two-way mixed-effects model, consistency type, assuming a fixed measurement system (AI-based analysis) and random participants.

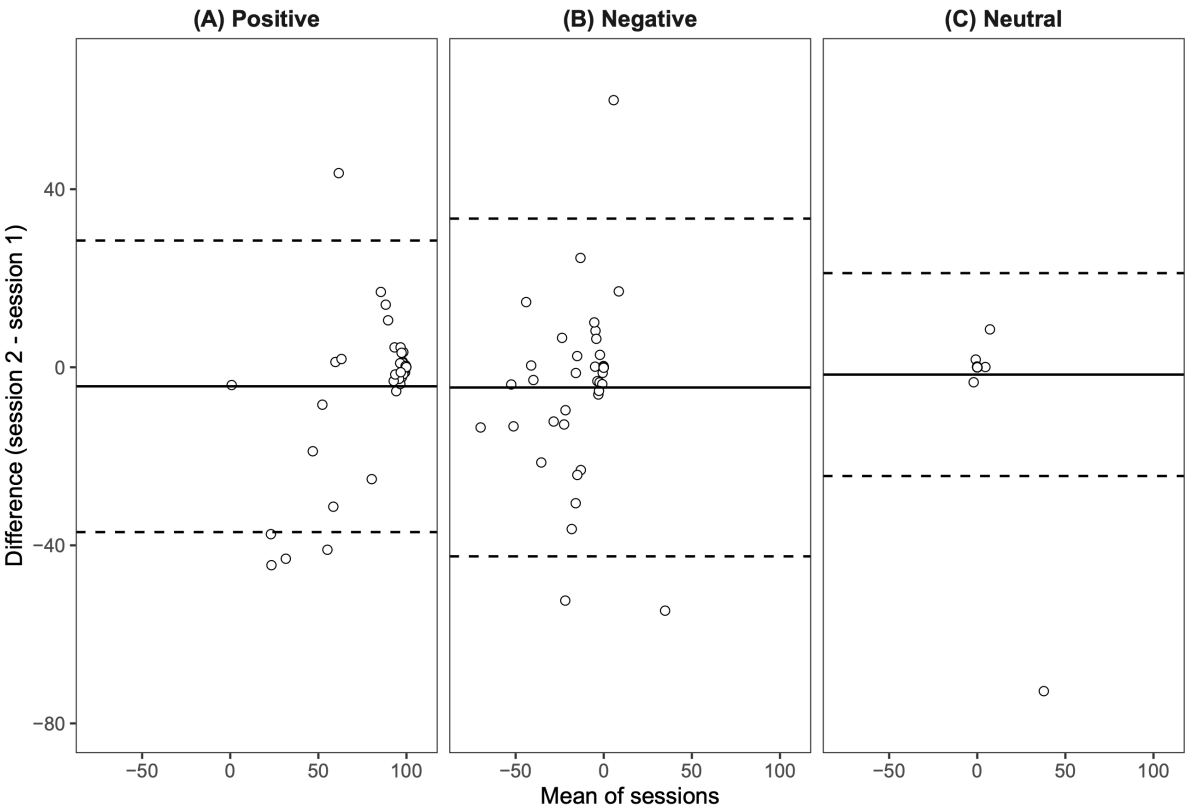


Figure 1 • Bland–Altman plots for the positive (A), negative (B), and neutral (C) conditions. Each circle represents one participant, plotted according to the mean of the two sessions on the x-axis and the difference between session 2 and session 1 on the y-axis. The solid lines represent the mean differences, and the dashed lines represent the 95% limits of agreement (mean ± 1.96 standard deviations). Across conditions, the mean differences were visually small, suggesting no clear systematic bias, while the neutral condition showed a notably narrower spread of valence scores, reflecting limited between-subject variability.

Table 2 • Absolute reliability indices.

Condition	Within-subject SD	95% LOA	Mean difference
Positive	11.81	−37.03 to 28.47	−4.28
Negative	13.69	−42.48 to 33.40	−4.54
Neutral	8.22	−24.43 to 21.15	−1.64
Combined	11.41	−35.11 to 28.14	−3.49

SD: standard deviation; LOA: limit of agreement. Mean difference was calculated as session 2 minus session 1. In the present two-session design, the standard error of measurement was numerically identical to the within-subject SD.

To support interpretation of the instructed valence conditions, supplementary descriptive and condition-congruence analyses were also performed. Condition-specific descriptive statistics showed that positive values were the highest, neutral values were centered near zero, and negative values were the lowest in both sessions (Table S1). The full expected ordering, positive > neutral > negative, was observed in 70.0% of participants in session 1 and 75.0% in session 2 (Table S2). In addition, within-session differences across the three conditions were significant in both sessions (Friedman $\chi^2 = 71.85$, $p < 0.001$ for session 1; $\chi^2 = 70.18$, $p < 0.001$ for session 2), and all pairwise comparisons remained significant after Holm adjustment (Table S3). These supplementary findings support broad valence-level condition congruence, although emotion estimation accuracy was not directly evaluated.

3.3. Short-term trait-like stability

When all state-specific data were combined to examine short-term trait-like stability, the integrated dataset yielded an ICC (3,1) of 0.94 (95% CI: 0.92–0.96; **Table 1**), indicating high short-term trait-like stability. This finding supports the presence of relatively consistent individual-level valence patterns across conditions over a short interval, beyond state-specific variation alone.

3.4. Sex-stratified sensitivity analysis

As a supplementary sensitivity analysis, state-specific reproducibility and short-term trait-like stability were also examined separately for male and female participants (Table S4). In male participants, the ICC (3,1) values were 0.87 for the positive condition, 0.61 for the negative condition, 0.48 for the neutral condition, and 0.96 for the combined dataset. In female participants, the corresponding ICC (3,1) values were 0.75, 0.29, 0.04, and 0.90, respectively. Although subgroup estimates showed some variation, the overall pattern was broadly consistent with the main analysis, with higher state-specific reproducibility in the positive condition, lower reproducibility in the neutral condition, and high short-term trait-like stability in the combined dataset. Because the subgroup sample sizes were limited, these findings should be interpreted as descriptive sensitivity analyses rather than formal evidence of sex differences.

4. Discussion

4.1. Summary of key findings

This study examined the reproducibility of AI-based valence for positive, negative, and neutral facial expression conditions in

healthy young adults in Japan. The ICC (3,1) was good for positive expressions, moderate for negative expressions, and poor for neutral expressions. The Bland–Altman analysis revealed that the mean differences between sessions were small in all conditions, and no clear systematic bias was observed. When data from all conditions were integrated, the resulting ICC was high, suggesting short-term trait-like stability at the individual level. Taken together, these findings suggest that the AI-based valence index may reflect both condition-specific variation and relatively stable individual-level patterns across emotional conditions.

4.2. Interpretation of state-specific valence reproducibility

The differences in reproducibility across conditions can be explained by the clarity and intensity of the facial cues associated with each expression. Positive facial expressions tend to evoke more distinct and easily identifiable facial movements, which is consistent with the finding that happy facial expressions are recognized more quickly and accurately than other emotions [25]. Negative facial expressions vary greatly from person to person, and the strength of the expression is not consistent, and this may have resulted in moderate reproducibility. Previous studies have shown that non-happy facial expressions generally have greater perceptual variability and weaker prototypicality [25, 26], resulting in less consistency in facial movement patterns. A meta-analysis reported an in-group advantage in emotion recognition and suggested that greater exposure between cultural groups attenuates this advantage, indicating that familiarity with culture-specific expressive styles can shape the perception of facial emotions [27]. In the present study, such familiarity may have contributed to differences in variability across valence conditions. However, because we did not directly assess participants’ everyday exposure to specific facial expressions, this interpretation should be considered tentative. In addition, the task instruction for the negative condition was broader than that for the positive condition, as participants were given multiple examples, including sad, frustrated, and disgusted facial expressions, which may have further increased heterogeneity in facial responses. As a descriptive sensitivity analysis, reproducibility was also examined separately by sex. Although the overall pattern was broadly similar between male and female participants across conditions, including the integrated dataset, the negative condition showed lower reproducibility in female than in male participants, with the female subgroup showing an ICC in the poor range. Although this finding should be interpreted cautiously because of the limited subgroup sample sizes, it may reflect sex-related variation in expressive consistency for negative facial expressions [28, 29]. Prior studies have suggested that sex differences in facial expressivity are not

uniform across emotions [29] and may be shaped by socialization and contextual influences [28], as well as by cultural display rules, including patterns observed in studies involving Japanese participants [30], although the direction and magnitude of these effects have not always been consistent.

On the other hand, the reproducibility of neutral facial expressions was clearly low, possibly because the range of valence scores in the neutral condition was quite narrow, resulting in little inter-subject variability. Because the ICC is highly dependent on score range and between-subject variance [13, 14], the very low ICC in the neutral condition should be interpreted with caution. In the present study, the neutral condition showed the smallest within-subject SD (8.22) and the narrowest 95% LOA (−24.43 to 21.15) among the three state-specific conditions. These findings indicate that absolute agreement between sessions was relatively stable, even though the ICC was markedly attenuated by restricted between-subject variability. Therefore, the low ICC in the neutral condition is better understood as a statistical consequence of the limited score dispersion rather than as evidence of poor within-subject reproducibility.

4.3. Interpretation of short-term trait-like stability

The high ICC obtained when data from all conditions were integrated suggests short-term trait-like stability at the individual level. However, this finding should be interpreted cautiously because the two recording sessions were separated by only 30 min; thus, the present design does not support claims about enduring personality-level affect. Rather, the integrated dataset may capture relatively stable individual-level patterns over a short time interval, rather than merely reflecting condition-specific fluctuations. In psychological research, both temporary state components and more stable trait-like components are considered important [8, 20]. Trait-like tendencies may contribute to relatively stable individual differences in emotional style and reactivity [7, 8]. The high ICCs in the integrated dataset may reflect that, when multiple emotional conditions are considered together, stable between-subject differences become more apparent while condition-specific range restriction becomes less influential [13, 14]. Therefore, the present findings are better interpreted as evidence of short-term stability in broad valence-related individual patterns across the specific experimental conditions used in this study, rather than as evidence of enduring trait affect.

4.4. Implications for digital environments

Digital environments increasingly mediate everyday social interactions and mental health support through video calls, online learning, telemedicine, and related platforms. In these contexts, facial information is acquired through cameras under a wide range of practical conditions. AI-based valence measures may therefore have potential as low-burden digital indicators for repeated emotional assessment and remote monitoring, provided that their measurement properties are adequately established [31, 32]. In the present study, integrated valence data across emotional conditions yielded a high ICC, indicating short-term trait-like stability and suggesting that AI-based valence may capture relatively stable individual-level patterns over short intervals. This characteristic may be useful for repeated assessment in digital environments. At the same time, digital environments introduce additional sources

of measurement variability, including camera viewpoint and distance, lighting, video quality, and task and social contexts, such as speaking versus listening and one-to-one versus group interaction. Therefore, future research should examine the effects of different devices, recording settings, and interaction contexts on measurement variability. In particular, reproducibility should be evaluated using ecologically valid real-world data obtained under in-the-wild conditions across multiple contexts [33].

4.5. Potential relevance for digital emotional assessment in East Asia

The findings of this study also have broader implications for mental health systems in East Asia. Across East Asia and related regional contexts, the availability and structure of mental health services vary widely, and many countries continue to lack sufficient community-based care and mechanisms for continuous monitoring of psychological well-being outside clinical settings [34]. According to the *WHO Mental Health Atlas 2024* [35], persistent shortages of mental health professionals, urban–rural disparities in service provision, limitations in community services, and substantial treatment gaps remain major challenges across the Southeast Asia and Western Pacific regions. Under such circumstances, low-burden and noninvasive approaches to emotional assessment may serve as useful adjuncts to existing service structures.

The present findings also suggest that AI-derived valence indices offer an initial foundation for low-burden emotional assessment in schools, communities, or other nonspecialist settings. Cultural factors remain important in this interpretation. In East Asian contexts, negative emotions are often expressed less openly, and culturally shaped display rules may influence observable facial behavior over time [36]. A valence index showing relative stability across multiple emotional conditions may therefore help support repeated emotional assessment while reducing overreliance on a single expression context. Although the present study included only healthy young adults, the findings provide an initial foundation for future community-oriented mental health research in East Asia.

4.6. Human factors in emotion production and recognition and their implications for AI-derived valence

In addition to the condition-specific interpretations discussed above, the present findings may also be understood in light of broader human variability in facial emotion production and recognition. Facial emotion processing involves multiple related but distinct components, including the production of facial expressions, the recognition of those expressions, and the interpretation of their underlying emotional states. These processes involve different mechanisms and should therefore be considered separately when interpreting the present findings. In particular, when posed facial expressions are used as experimental stimuli, they may not necessarily reflect the expressor's internal emotional state and may be more appropriately understood as reflecting perceiver responses to outward facial appearance [37]. This distinction is important for the interpretation of the present findings. From the perspective of expression production, individuals differ in their ability to produce posed emotional expressions, and such expressions are influenced not only by emotional content but

also by voluntary control and task instructions and experimental constraints [37, 38]. An experimental study has demonstrated that facial emotion production can be assessed as an ability, and that performance varies across individuals and across emotion categories [38]. In addition, the accuracy of imitating facial expressions differs by emotion type, with positive expressions generally being easier to reproduce than several negative expressions [39].

From the perspective of emotion recognition, human observers also show variability in accuracy across emotion categories, with some expressions being recognized more reliably than others. In studies of healthy adults, happiness tends to be recognized with higher accuracy and shorter decision times than several negative expressions, whereas negative expressions such as fear, disgust, and sadness show higher rates of misclassification [39, 40]. Recognition performance is further influenced by perceptual strategies, including the exploration of emotion-specific diagnostic facial regions, with more frequent exploration of these regions being associated with faster correct recognition [41]. Neutral expressions present an additional challenge. Neutral faces are ambiguous and are often misjudged or interpreted with a negative bias, with participants tending to perceive them as sad [42]. This has implications for the interpretation of the present findings, as the markedly low reproducibility observed in the neutral condition is likely attributable to restricted variance, but may also partly reflect the limited stability of neutral facial expressions as emotional signals.

Taken together, these findings indicate that human production and recognition of facial emotional expressions exhibit systematic variability. Accordingly, the pattern observed in the present study—high reproducibility in the positive condition, moderate reproducibility in the negative condition, and markedly low reproducibility in the neutral condition—should be interpreted within the broader context of human facial emotion production and recognition. These findings suggest that the observed variability in AI-derived valence scores may not be specific to the AI system alone, but may also be associated with systematic variability in human facial emotion production and recognition.

4.7. Limitations and future directions

This study has several limitations. First, the sample consisted of only healthy young adults, and the findings should not be generalized to other age groups or clinical populations. Second, this study used directed, posed facial expressions rather than spontaneous, context-dependent ones. Although this approach was necessary to standardize the recording conditions across participants, posed expressions do not necessarily reflect the expressor's internal emotional state and do not fully reflect the complexity, variability, and social context of real-world emotional behavior. Accordingly, the generalizability of the findings to ecologically valid real-world settings, including naturalistic digital monitoring environments, remains limited. Future studies should examine AI-based valence reproducibility using more naturalistic emotional responses, for example, through emotion induction paradigms based on short movie clips or socially contextualized tasks. Third, the range of valence values in the neutral condition was extremely narrow, and the resulting restriction of between-subject variability substantially attenuated the ICC. This reflects a methodological property of ICC, which is strongly influenced by score dispersion and between-subject variance, rather than necessarily indicating poor

within-subject reproducibility. In the present study, the neutral condition showed relatively small within-subject SDs and LOA, suggesting that absolute agreement between sessions remained comparatively stable despite the low ICC. Additionally, the recordings were collected in a controlled environment, and their reproducibility in real-world digital environments, such as video calls and mobile recordings, has yet to be established. Thus, ICC-based reproducibility estimates should be interpreted cautiously when evaluated in a single condition with low variability in affective states. The high ICC observed in the integrated dataset suggests that combining multiple conditions may help reveal short-term individual-level stability, although this should not be interpreted as evidence of enduring trait affect. In addition, this study was designed to evaluate the reproducibility of AI-derived valence scores rather than the accuracy of fine-grained discrete emotion classification. Although the participants were instructed to pose positive, neutral, and negative facial expressions, the accuracy with which they expressed specific target emotions and the accuracy with which the AI classified those expressions into discrete emotional categories were not directly evaluated. Because the Kokoro Sensor is a proprietary system and its internal feature extraction and valence computation processes are not publicly available, we could not directly evaluate whether positive and negative valence estimates were based on structurally comparable Action Unit-related feature sets. Accordingly, the interpretation that positive expressions may be more prototypical, whereas negative expressions may be more heterogeneous, should be regarded as an indirect hypothesis rather than a directly tested mechanism. Therefore, the present findings should be interpreted as evidence of the reproducibility of broad valence estimates rather than fine-grained emotion recognition accuracy. Finally, this study focused on test-retest reproducibility; other psychometric properties, such as sensitivity to change, criterion-related validity, and cross-cultural generalizability, were not examined. Further research is needed to clarify these points.

5. Conclusions

The present study examined the reproducibility of AI-derived emotional valence values in healthy young adults across positive, negative, and neutral facial expression conditions. Reproducibility differed across conditions, with positive and negative expressions showing good to moderate test-retest reliability, while the neutral condition showed low ICC values as a result of restricted score variability. When valence data from all conditions were integrated, the resulting ICC was high, indicating short-term trait-like stability at the individual level rather than purely condition-specific fluctuation. These findings support the potential of AI-based valence as a low-burden measure for future emotional assessment in digital environments. However, further validation under ecologically valid real-world conditions is needed.

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Author contributions

Conceptualization, S.A., S.Y. and K.T.; methodology, S.A., S.Y. and K.T.; software, S.A.; validation, S.A., S.Y. and K.S.; formal analysis, S.A.; investigation, S.A. and S.Y.; resources, S.A. and K.T.; data curation, S.A. and S.Y.; writing—original draft preparation, S.A.; writing—review and editing, S.A., S.Y., K.S. and K.T.; visualization, S.A.; supervision, K.T.; project administration, S.A.; funding acquisition, S.A. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare that they have no competing interests.

Data availability statement

The data supporting the findings of this publication can be made available upon request.

Institutional review board statement

The study was conducted according to the guidelines of the Declaration of Helsinki and approved by the ethics committee of the School of Allied Health Sciences, Kitasato University, Japan (Approval No. 2023-032; approved on 19 January 2024).

Informed consent statement

Informed consent for participation was obtained from all participants involved in the study.

Supplementary materials

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